Neo and SuperNeo

The prover commits to a CCS witness using a lattice-based commitment scheme with pay-per-bit costs, runs a single invocation of the sum-check protocol over an extension of a small prime field, and achieves plausible post-quantum security under a standard structured lattice assumption (Module-SIS).

Key idea

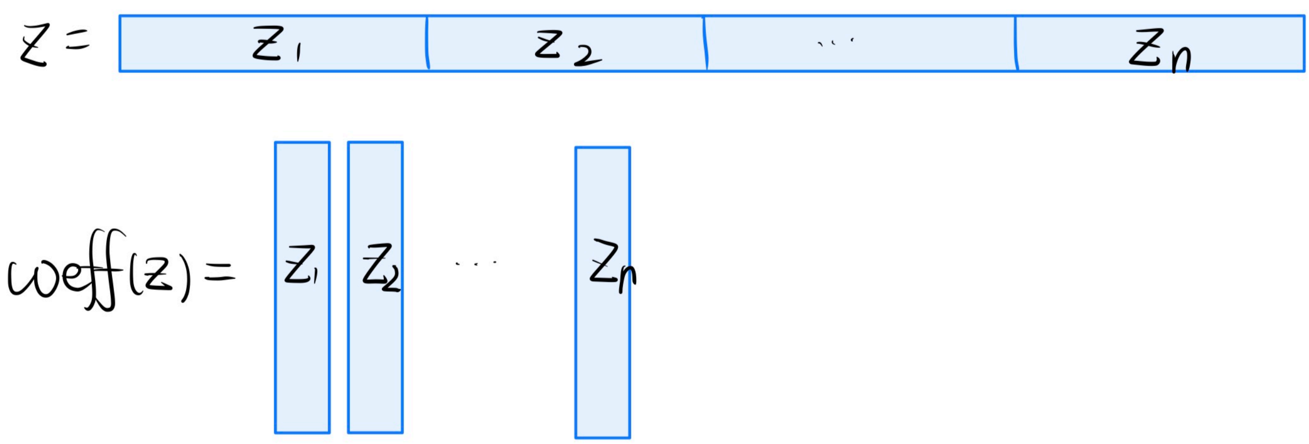
- sumcheck + normcheck over the finite field (Running sumcheck over rings is expensive)
- Commit the witness using Ajtai over ring
- from Fq to Rq: coefficient embedding
- keep linear evaluation: inner-product

Backgrounds

1. Coefficient Embedding (SuperNeo Embedding)

Given a field vector $z \in \mathbb{F}^{d \cdot n}$, split the vector into n sub-vectors of length d each, i.e., $z = [z_1, z_2, \ldots, z_n]$ where each $z_i := [z_{i,1}, \ldots, z_{i,d}] \in \mathbb{F}^d$. We will embed each sub-vector z_i as the coefficients of a single ring element $z_i := \sum_j z_{i,j} X^{j-1} \in R_{\mathbb{F}}$. The resulting vector of ring elements $z := [z_1, z_2, \ldots, z_n] \in R_{\mathbb{F}}^n$ is the SuperNeo embedding of z.

$$z = [\;\; z_1 \;\;|\;\; z_2 \;\;|\;\; \cdots \;\;|\;\; z_n \;\;]$$



2. Linear Relation: $\mathbb{F} \to \mathcal{R}$

Inner Product Transform.

There exists a linear transform $\bar{\cdot} : \mathbb{F}^d \to \mathbb{F}^d$ such that for all $a, b \in \mathbb{F}^d$, we have the constant term

$$\mathrm{ct}(\bar{a} \cdot b) = \langle a, b \rangle$$

where $\bar{a}$ denotes applying the transform to a and embedding $\bar{a}$ into the ring.

Example.

Take $\mathcal{R} = \mathbb{Z}[X]/(X^d + 1)$ as an example, such a transform is $\bar{a}(X) = a(X^{-1})$, i.e., evaluate at X^{-1}. In this ring, $-1 = X^d, X^{-1} = -X^{d-1}, X^{-2} = -X^{d-2}$...

> Let $a = [1, 2, 3, 4]$, $b = [5, 6, 7, 8] \in \mathbb{F}^4$, with $\mathcal{R} = \mathbb{Z}[X]/(X^4 + 1)$.

> Embed as polynomials: $a(X) = 1 + 2X + 3X^2 + 4X^3, \;\; b(X) = 5 + 6X + 7X^2 + 8X^3$.

> In this ring $X^{-1} = -X^3$,
>
> $$\bar{a}(X) = 1 + 2X^{-1} + 3X^{-2} + 4X^{-3}$$
> $$= 1 + 2(-X^3) + 3(-X^2) + 4(-X) = 1 - 4X - 3X^2 - 2X^3.$$

> The constant term of $\bar{a}(X) \cdot b(X)$ collects coefficients $\bar{a}_i \cdot b_j$ where $i + j \equiv 0 \pmod 4$ with appropriate sign from $X^4 \equiv -1$:
>
> $$\mathrm{ct}(\bar{a} \cdot b) = \bar{a}_0 b_0 - \bar{a}_1 b_3 - \bar{a}_2 b_2 - \bar{a}_3 b_1$$
> $$= (1)(5) - (-4)(8) - (-3)(7) - (-2)(6) = 1 \times 5 + 4 \times 8 + 3 \times 7 + 2 \times 6.$$

Linear Relations.

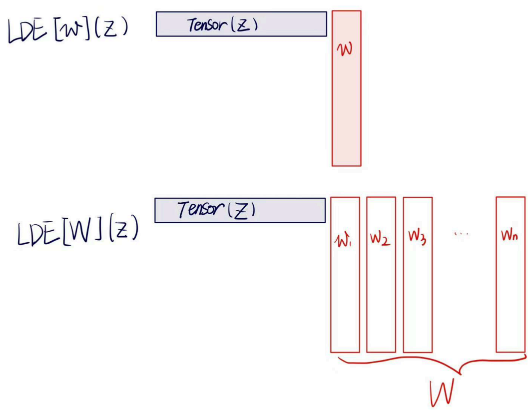
For a matrix $A \in \mathbb{F}^{m \times (d \cdot n)}$ and a witness vector $z \in \mathbb{F}^{d \cdot n}$. After embedding z into ring elements $\boldsymbol{z} \in R_{\mathbb{F}}^n$, each row $a_i \in \mathbb{F}^{d \cdot n}$ of A can similarly be split and embedded. Then the i-th entry of the product is:

$$[Az]_i = \langle a_i, z \rangle = \mathrm{ct}(\bar{\boldsymbol{a}}_i \cdot \boldsymbol{z})$$

where $\bar{\boldsymbol{a}}_i \in R_{\mathbb{F}}^n$ is the transformed embedding of row a_i. In matrix form, defining $\bar{\boldsymbol{A}} \in R_{\mathbb{F}}^{m \times n}$ as the matrix of transformed row embeddings:

$$Az = \mathrm{ct}(\bar{\boldsymbol{A}} \cdot \boldsymbol{z}) \in \mathbb{F}^m$$

where $\mathrm{ct}(\cdot)$ is applied entry-wise. This means a *single ring matrix-vector multiplication* $\bar{\boldsymbol{A}} \cdot \boldsymbol{z}$ over $R_{\mathbb{F}}^n$ encodes all m linear constraints simultaneously — the constant term extracts the result back into $\mathbb{F}^m$.



Relations

1. Structure: is a collection of elements:

$$\mathsf{s} := \big\{\{M_j \in \mathbb{F}^{m\times n_\mathbb{F}}\}_{j\in[t]}, f \in \mathbb{F}^{<u}[X_1,\ldots,X_t]\big\},$$

which consists of *matrices and a degree-u polynomial*.

2. Norm-bounded CCS

$\mathcal{L}$: Ajtai commitment over $\mathcal{R}$, $(applyFFT)$.

Let $\mathcal{L} : \mathbf{R}_\mathbb{F}^{n_\mathbf{R}} \to \mathbb{C}$ be an arbitrary $\mathbf{R}_\mathbb{F}$-module homomorphism. Let s be a structure.

$$\mathrm{CCS}(b,\mathcal{L}) = \left\{\begin{array}{l} \mathsf{s};\, \big(c \in \mathbb{C}, x \in \mathbb{F}^{n_{\mathbb{F},\mathrm{in}}}; w \in \mathbb{F}^{n_\mathbb{F}-n_{\mathbb{F},\mathrm{in}}}\big) : \\ \quad c = \mathcal{L}(z) \wedge \|z\|_\infty < b \wedge \\ \quad f(\widetilde{M_1 z},\ldots,\widetilde{M_t z}) \in \mathbf{ZS}_{\log m} \end{array}\right\}$$

where $z := [x, w]$, $\mathbf{ZS}_{\log m}$ is a 'zero set' st $f(x) = 0$ at all $\{0,1\}^{\log m}$.

3. Norm-bounded CCS Evaluation Relation.

Let s be a structure. Let $\mathcal{L} : \mathbf{R}_\mathbb{F}^{n_\mathbf{R}} \to \mathbb{C}$ be an arbitrary $\mathbf{R}_\mathbb{F}$-module homomorphism. Define $\mathcal{L}_\mathrm{in} : \mathbf{R}_\mathbb{F}^{n_\mathbf{R}} \to \mathbf{R}_\mathbb{F}^{n_{\mathbf{R},\mathrm{in}}}$ to be the trivial $\mathbf{R}_\mathbb{F}$-module homomorphism that projects the first $n_{\mathbf{R},\mathrm{in}}$ indices in columns.

$$\mathrm{CE}(b,\mathcal{L}) = \left\{\begin{array}{l} \left(\left(\mathsf{s}; \left(\begin{array}{c} c \in \mathbb{C}, \\ x \in \mathbb{F}^{n_{\mathbb{F},\mathrm{in}}}, \\ r \in \mathbb{K}^{\log m}, \\ \{y_j \in \mathbf{R}_\mathbb{K}\}_{j\in[t]} \end{array}\right)\right); z \in \mathbb{F}^n\right) : \\ \quad c = \mathcal{L}(z) \wedge x = \mathcal{L}_\mathrm{in}(z) \wedge \\ \quad \|z\|_\infty < b \wedge \\ \quad \forall j \in [t], y_j = \widetilde{M_j z}(r) \end{array}\right\}$$

Schemes

At the beginning, CCS witnesses are decomposed into low-norm, i.e. $Norm-boundedCCS$.

Step 1. sumcheck

$\mathrm{F}(\vec{X})$ encodes the CCS constraints (all K of them). $\mathrm{NC}(\vec{X})$ encodes the norm constraints (all $K+k$ of them). $\mathrm{Eval}(\vec{X})$ encodes the evaluation claims (all k of them) from the prior step. Finally, $Q(\vec{X})$ is defined such that if its sum over the boolean hypercube $\{0,1\}^{\log(m)}$ equals to the constructed sum T, then all the respective checks hold.

Input $\in \mathrm{CCS}(b,\mathcal{L})^K \times \mathrm{CE}(b,\mathcal{L})^k$

$$\big(\mathsf{s};\quad (c_i \in \mathbb{C},\ x_i \in \mathbb{F}^{n_{\mathbb{F},\mathrm{in}}});\ w_i \in \mathbb{F}^{n_\mathbb{F}-n_{\mathbb{F},\mathrm{in}}}\big)_{i=1}^{K},$$

$$\big(\mathsf{s};\quad c_i \in \mathbb{C},\ x_i \in \mathbb{F}^{n_{\mathbb{F},\mathrm{in}}},\ r \in \mathbb{K}^{\log m},\ \{y_{i,j} \in \mathbf{R}_\mathbb{K}\}_{j\in[t]};\ z_i \in \mathbb{F}^{n_\mathbb{F}}\big)_{i=K+1}^{K+k}$$

Output $\in \mathrm{CE}(b,\mathcal{L})^{K+k}$

$$\big(\mathsf{s};\quad c_i \in \mathbb{C},\ x_i \in \mathbb{F}^{n_{\mathbb{F},\mathrm{in}}},\ r' \in \mathbb{K}^{\log m},\ \{y'_{i,j} \in \mathbf{R}_\mathbb{K}\}_{j\in[t]};\ z_i \in \mathbb{F}^{n_\mathbb{F}}\big)_{i\in[K+k]}$$

Setup $\mathcal{G}(1^\lambda, n_{\mathrm{R}}) \to \mathsf{pp}$: Output $\mathsf{pp} \leftarrow \mathsf{Setup}(1^\lambda, n_{\mathrm{R}})$.

Encoder $\mathcal{K}(\mathsf{pp}, \mathsf{s}) \to (\mathsf{pk}, \mathsf{vk})$: Output $\big((\mathsf{pp}, \mathsf{s}), \perp\big)$.

Reduction $\langle \mathcal{P}, \mathcal{V} \rangle((\mathsf{pk}, \mathsf{vk}), u_1, w_1) \to (u_2; w_2)$:

1. $\mathcal{V}$: Send challenges $\alpha \xleftarrow{\$} \mathbb{K}^{\log m}$ and $\gamma \xleftarrow{\$} \mathbb{K}$ to $\mathcal{P}$.
2. $\mathcal{V} \leftrightarrow \mathcal{P}$: For all $i \in [K]$, define $z_i := [x_i,\ w_i]$. Define $\vec{X} := (X_1, \ldots, X_{\log m})$,

$$\mathsf{F}(\vec{X}) := \sum\nolimits_{i=1}^{K} \gamma^{i-1} \cdot f\Big(\widetilde{M_1 z_i},\ \ldots,\ \widetilde{M_t z_i}\Big) \in \mathbb{K}[\vec{X}]$$

$$\mathsf{NC}(\vec{X}) := \sum\nolimits_{i=1}^{K+k} \gamma^{i-1} \cdot \prod\nolimits_{j=-b-1}^{b-1}\Big(\widetilde{z_i} - j\Big) \in \mathbb{K}[\vec{X}]$$

$$\mathsf{Eval}(\vec{X}) := \mathsf{eq}\big(\vec{X},\ r\big) \cdot \sum_{i=K+1}^{K+k} \sum_{j=1}^{t} \sum_{\ell=1}^{d} \gamma^{\mathrm{I}(i,j,\ell)} \cdot \widetilde{\mathsf{cf}\big(\bar{\boldsymbol{M}}_j \boldsymbol{z}_i\big)_\ell} \in \mathbb{K}[\vec{X}]$$

   where $\mathrm{I}(i,j,\ell) = \big(i\text{-}(K+1)\big) + k(j\text{-}1) + kt(\ell\text{-}1)$ and $\widetilde{\mathsf{cf}\big(\bar{\boldsymbol{M}}_j \boldsymbol{z}_i\big)_\ell}$ is the multilinear extension of the ℓ-th coefficient vector of $\bar{\boldsymbol{M}}_j \boldsymbol{z}_i$ (Definition 2). Define

$$Q(\vec{X}) := \mathsf{eq}(\vec{X},\ \alpha) \cdot \Big(\mathsf{F}(\vec{X}) + \gamma^{K} \cdot \mathsf{NC}(\vec{X})\Big) + \gamma^{2K+k} \cdot \mathsf{Eval}(\vec{X}) \in \mathbb{K}[\vec{X}]$$

   Define claimed sum of Q over $\{0,1\}^{\log m}$ as

$$\mathrm{T} := \sum_{i=K+1}^{K+k} \sum_{j=1}^{t} \sum_{\ell=1}^{d} \gamma^{\mathrm{I}(i,j,\ell)} \cdot \mathsf{cf}(y_{i,j})_\ell \in \mathbb{K}$$

   Perform $\mathsf{SumCheck}$ $(\mathrm{T};\ Q)$ (Definition 6) which reduces the claim that

$$\mathrm{T} = \sum\nolimits_{\vec{x} \in \{0,1\}^{\log m}} Q(\vec{x})$$

   to a new evaluation claim $v \stackrel{?}{=} Q(r')$ for new evaluation point $r' \in \mathbb{K}^{\log m}$.
3. $\mathcal{P}$: Send $\forall\, i \in [K+k]$, $\forall\, j \in [t]$, $y'_{i,j} \leftarrow \widetilde{\bar{\boldsymbol{M}}_j \boldsymbol{z}_i}(r') \in \mathrm{R}_{\mathbb{K}}$.
4. $\mathcal{V}$: Derive the claimed intermediate evaluations (Remark 2),

$$\mathrm{F} := \sum\nolimits_{i=1}^{K} \gamma^{i-1} \cdot f\big(\mathsf{ct}(y'_{i,1}),\ \ldots,\ \mathsf{ct}(y'_{i,t})\big) \in \mathbb{K}$$

$$\mathrm{N} := \sum\nolimits_{i=1}^{K+k} \gamma^{i-1} \cdot \prod\nolimits_{j=-b-1}^{b-1}\big(\mathsf{ct}(y'_{i,1}) - j\big) \in \mathbb{K}$$

$$\mathrm{E} := \mathsf{eq}(r',\, r) \sum_{i=K+1}^{K+k} \sum_{j=1}^{t} \sum_{\ell=1}^{d} \gamma^{\mathrm{I}(i,j,\ell)} \cdot \mathsf{cf}\left(y'_{i,j}\right)_{\ell} \in \mathbb{K}$$

Check the evaluation claim $v \stackrel{?}{=} Q\left(r'\right)$ as follows,

$$v \stackrel{?}{=} \mathsf{eq}(r',\, \alpha) \cdot \left(\mathrm{F} + \gamma^{K} \cdot \mathrm{N}\right) + \gamma^{2K+k} \cdot \mathrm{E}$$

5. Output $\left(\mathsf{s};\ c_i,\ x_i,\ r',\ \{y'_{i,j}\}_{j\in[t]};\ z_i\right)_{i\in[K+k]}$

Step 2. fold

Random linear combination reduction Π_{RLC}

Parameters: Refer to Definition 14.

Input $\in \mathsf{CE}(b, \mathcal{L})^{K+k}$

$\left(\mathsf{s};\ \ c_i \in \mathbb{C},\, x_i \in \mathbb{F}^{n_{\mathbb{F},\mathsf{in}}},\, r \in \mathbb{K}^{\log m},\, \{y_{i,j} \in \mathrm{R}_{\mathbb{K}}\}_{j\in[t]};\ \ z_i \in \mathbb{F}^{n_{\mathbb{F}}}\right)_{i\in[K+k]}$

Output $\in \mathsf{CE}(B, \mathcal{L})$

$\left(\mathsf{s};\ \ c \in \mathbb{C},\, x \in \mathbb{F}^{n_{\mathbb{F},\mathsf{in}}},\, r \in \mathbb{K}^{\log m},\, \{y_j \in \mathrm{R}_{\mathbb{K}}\}_{j\in[t]};\ \ z \in \mathbb{F}^{n_{\mathbb{F}}}\right)$

Setup $\mathcal{G}(1^\lambda, n_{\mathrm{R}}) \to \mathsf{pp}$: Output $\mathsf{pp} \leftarrow \mathsf{Setup}(1^\lambda, n_{\mathrm{R}})$.

Encoder $\mathcal{K}(\mathsf{pp}, \mathsf{s}) \to (\mathsf{pk}, \mathsf{vk})$: Output $\big((\mathsf{pp}, \mathsf{s}), \perp\big)$.

Reduction $\langle \mathcal{P}, \mathcal{V}\rangle((\mathsf{pk}, \mathsf{vk}), u_1, w_1) \to (u_2; w_2)$:

1. $\mathcal{V}$: Sample $\rho_1, \ldots, \rho_{K+k} \stackrel{\$}{\leftarrow} \mathcal{C}$ and compute:

$$c \leftarrow \sum_{i\in[K+k]} \rho_i c_i, \quad \boldsymbol{x} \leftarrow \sum_{i\in[K+k]} \rho_i \boldsymbol{x}_i, \quad \forall j \in [t],\ y_j \leftarrow \sum_{i\in[K+k]} \rho_i y_{i,j}$$

   Send $\rho_1, \ldots, \rho_\ell$ to $\mathcal{P}$.

2. $\mathcal{P}$: Compute $\boldsymbol{z} \leftarrow \sum_{i\in[K+k]} \rho_i \boldsymbol{z}_i$.

$\mathcal{C}$ is a subtractive set over $\mathcal{R}$

Step 3. decompose

Decomposition reduction Π_{DEC}

Parameters: Refer to Definition 14.

Input $\in \mathsf{CE}(B, \mathcal{L})$

$\big(\mathsf{s};\quad c \in \mathbb{C},\ x \in \mathbb{F}^{n_{\mathbb{F},\mathsf{in}}},\ r \in \mathbb{K}^{\log m},\ \{y_j \in \mathrm{R}_{\mathbb{K}}\}_{j\in[t]};\quad z \in \mathbb{F}^{n_{\mathbb{F}}}\big)$

Output $\in \mathsf{CE}(b, \mathcal{L})^k$

$\big(\mathsf{s};\quad c_i \in \mathbb{C},\ x_i \in \mathbb{F}^{n_{\mathbb{F},\mathsf{in}}},\ r \in \mathbb{K}^{\log m},\ \{y_{i,j} \in \mathrm{R}_{\mathbb{K}}\}_{j\in[t]};\quad z_i \in \mathbb{F}^{n_{\mathbb{F}}}\big)_{i\in[k]}$

Setup $\mathcal{G}(1^\lambda, n_{\mathrm{R}}) \to \mathsf{pp}$: Output $\mathsf{pp} \leftarrow \mathsf{Setup}(1^\lambda, n_{\mathrm{R}})$.

Encoder $\mathcal{K}(\mathsf{pp}, \mathsf{s}) \to (\mathsf{pk}, \mathsf{vk})$: Output $\big((\mathsf{pp}, \mathsf{s}), \perp\big)$.

Reduction $\langle \mathcal{P}, \mathcal{V}\rangle((\mathsf{pk}, \mathsf{vk}), u_1, w_1) \to (u_2; w_2)$:

1. $\mathcal{P}$: Compute $\big(c_i,\ \{y_{i,j}\}_{j\in[t]};\ z_i\big)_{i\in[k]}$ as follows,

$$(z_1, \ldots, z_k) \leftarrow \mathsf{split}_b(z), \quad c_i \leftarrow \mathcal{L}(\boldsymbol{z}_i), \quad \forall j \in [t],\ y_{i,j} \leftarrow \widetilde{\bar{\boldsymbol{M}}_j \boldsymbol{z}_i}(r)$$

   Send $\big(c_i,\ \{y_{i,j}\}_{j\in[t]}\big)_{i\in[k]}$ to $\mathcal{V}$.

2. $\mathcal{V}$: Compute $(x_1, \ldots, x_k) \leftarrow \mathsf{split}_b(x)$. Check the following equations,

$$c \stackrel{?}{=} \sum\nolimits_{i\in[k]} b^{\,i-1} \cdot c_i \quad \text{and} \quad \forall j \in [t],\ y_j \stackrel{?}{=} \sum\nolimits_{i\in[k]} b^{\,i-1} \cdot y_{i,j}$$

   where the norm-bound b is treated as a field element.

3. Output $\big(\mathsf{s};\ c_i,\ x_i,\ r,\ \{y_{i,j}\}_{j\in[t]};\ z_i\big)_{i\in[k]}$

6 properties

- plausible post-quantum security
- pay-per-bit commitment costs:
  The cost to commit to a witness should scale with the bit-width of the witness values: for example, committing to a vector of b-bit values should be roughly b times cheaper than committing to values that span the full field.
  Group-based schemes achieve this via Pedersen or KZG commitments.
  Since the Ajtai commitment scheme requires decomposing these arbitrary-norm ring elements for binding, the commitment cost is the same whether the original values are 1-bit or 64-bit.
- field-native arithmetic (the sum-check and norm checks run purely over a small field):
  operate natively over a prime field (or extension field), without performing polynomial ring arithmetic.
- support for general (non-SIMD) constraint systems:
  support general NP-complete constraint systems such as CCS over a single witness vector, without requiring the constraint system to be “data parallel” (SIMD, Single Instruction, Multiple Data).

- small-field (64bits, fits within a machine register) support:
  work natively over small prime fields, including popular SNARK-friendly fields such as Goldilocks.
- low recursion overheads:
  The recursive verifier circuit should be small enough that the per-step prover cost of IVC remains practical. More broadly, folding schemes that rely on ring sum-check techniques —such as LatticeFold and SALSAA — inherit high recursion overheads because the recursive verifier must hash ring elements rather than field elements.